GAMEPLAYING
MINIMAX & ALPHA BETA CUTOFF

MINIMAX

- Minimax is a method used to evaluate game trees.
- A static evaluator is applied to leaf nodes, and values are passed back up the tree to determine the best score that can be obtained against a rational opponent.
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Minimax Principle

Assume that the opponent will choose the minimizing move next (after your move)

so you now *choose the best move under this assumption*

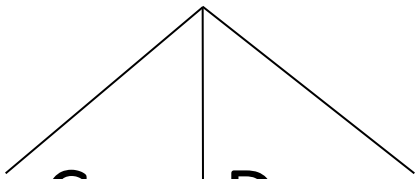
i.e., the maximum (highest-value) option considering **both** your move and the opponent's optimal move.

we can look ahead more than 2 moves: we can search ahead as far as we can afford.

One ply search

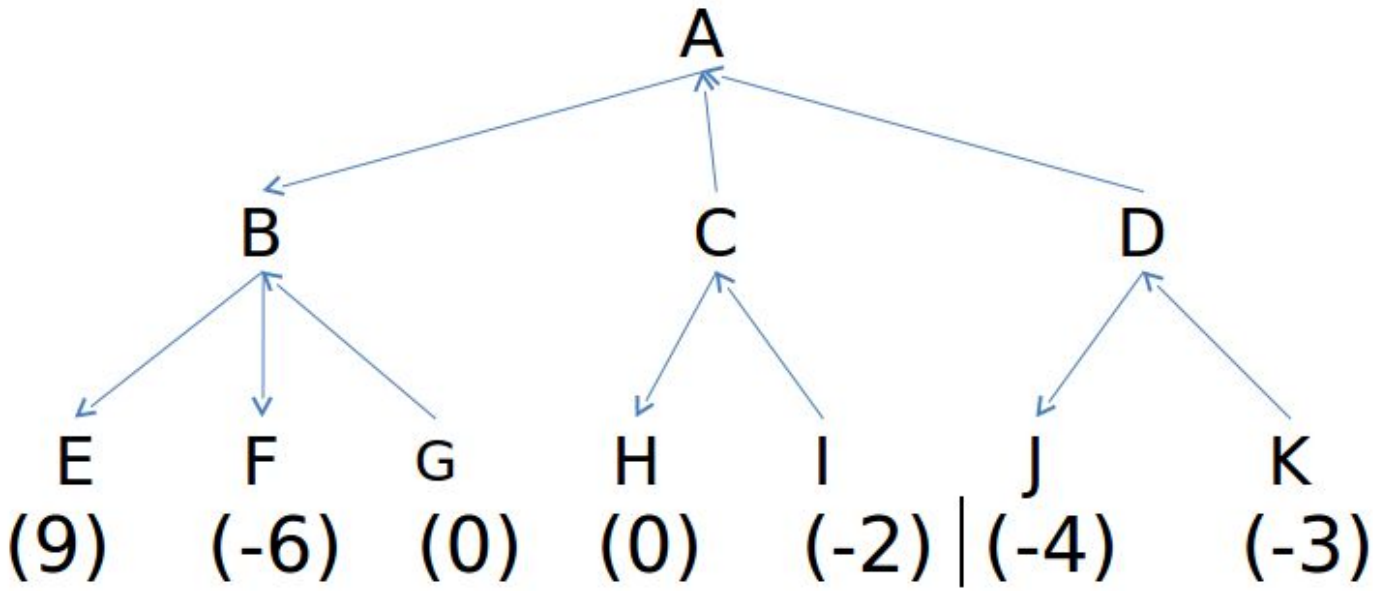
A will choose move to B.

A (maximising player)

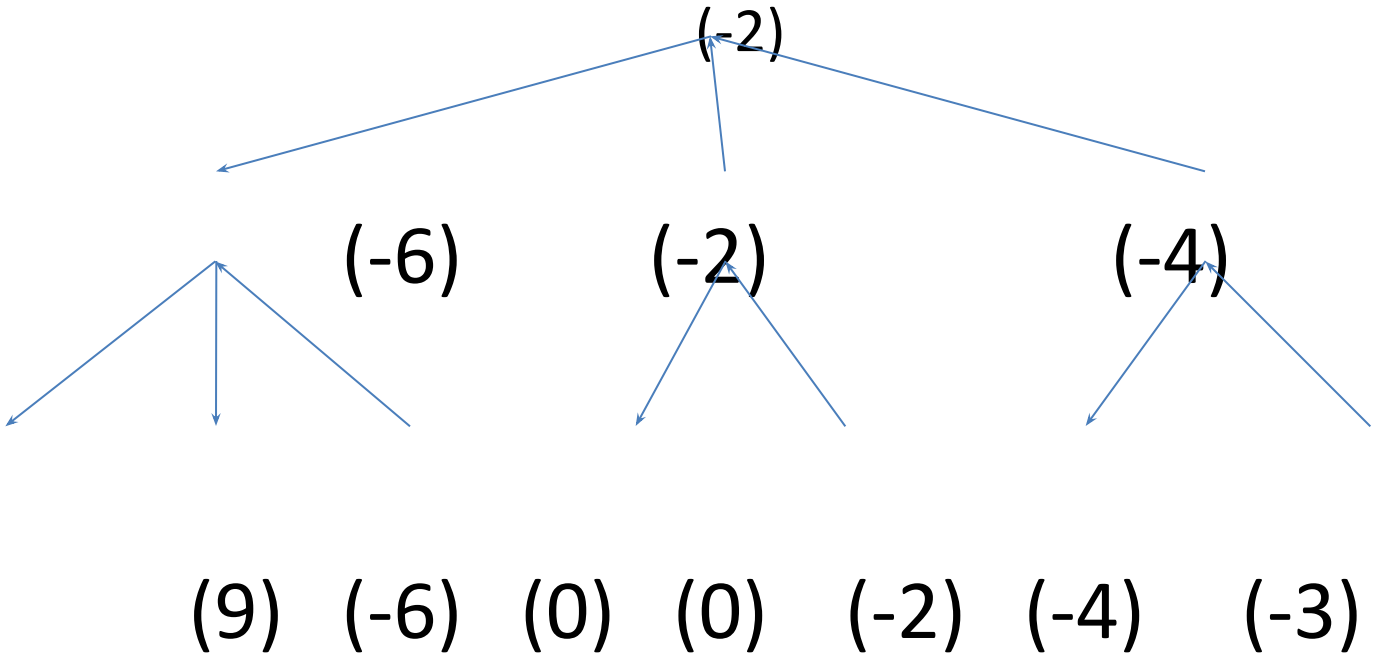


B	C	D
8	3	-2

TWO-PLY SEARCH



BACKING UP THE VALUES OF A TWO-PLY SEARCH



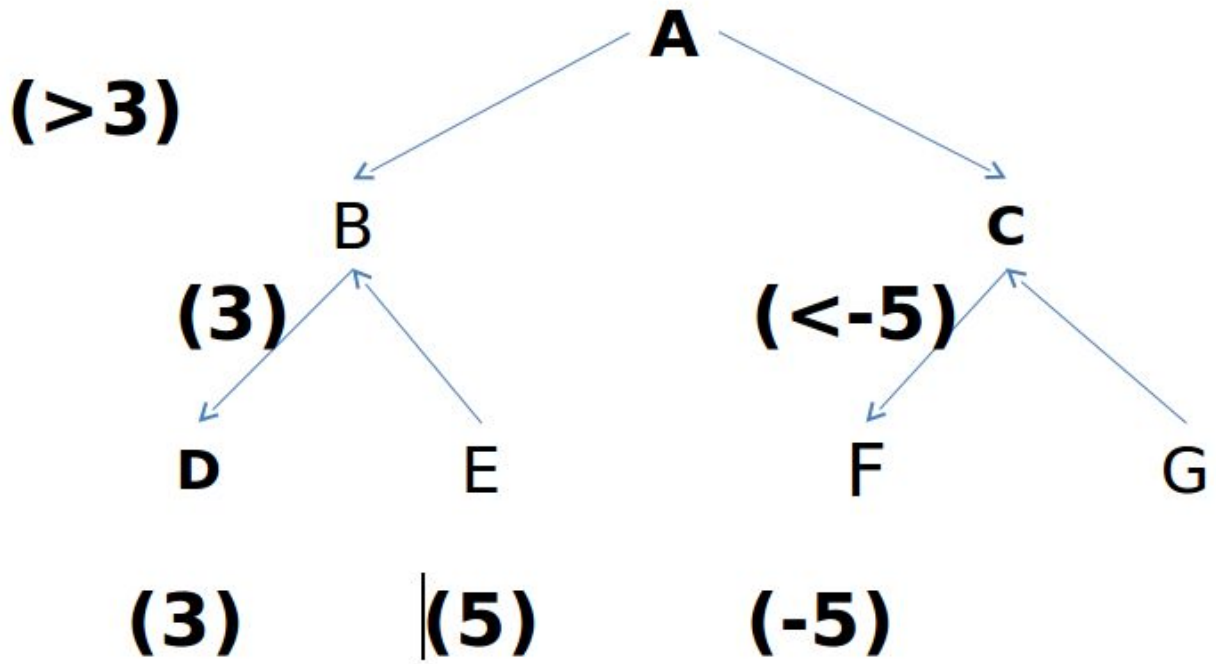
ALPHA BETA CUT-OFF

- The alpha-beta procedure can speed up a depth-first minimax search.

Alpha: a lower bound on the value that a max node may ultimately be assigned.

Beta: an upper bound on the value that a minimizing node may ultimately be assigned.

AN ALPHA CUT-OFF



Alpha beta cut off contd.....

Alpha value is used by a minimising node to cut down the search space.

Beta value is used by a maximising node to cut down the search space.